


**BITS PILANI K.K. BIRLA GOA CAMPUS**

**Department of Physics**

# What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

**Step 9: Mass Recovery and Fit Stability**

## Group-B

| | |
|---|---|
| Priyansh Rawat | 2023B5A70968G |
| Mayank Sinha | 2023B5A70942G |
| Samarth Bhatnagar | 2023B5A70853G |

Under Professor: Rudranil Basu

29 October 2026

# 1 Frequency profiling and mock-data calibration

We fit

$$F(r) = b_0 + b_2 r^2 + r^{3/2}[A\cos(\mu \ln r) + B\sin(\mu \ln r)]. \tag{1}$$

For each trial $\mu$, the remaining coefficients are linear. We scan $0.1 \le \mu \le 4$ in increments 0.01, refine every local minimum and retain competing solutions. Known Gaussian mock errors define a profile likelihood; the nominal 95% set is $\Delta\chi^2 \le 3.841459$, including disconnected intervals. The mocks use 50 log-spaced points on $10^{-3} \le r \le 0.2$, $M = 2$, $g = 0.1$, $A = 0.1$, $B = 0.05$, $b_0 = 0$, $b_2 = 0.2$. Their known frequency is $\sqrt{1.76}$. Independent noise has standard deviation 1% or 5% of the oscillatory RMS. Seed 20261029 gives 100 realisations at each noise level.

| Noise | Frequency bias | Frequency RMSE | 95% coverage |
|---|---|---|---|
| 1% | 0.002425 | 0.04687 | 93% |
| 5% | 0.02179 | 0.205 | 94% |

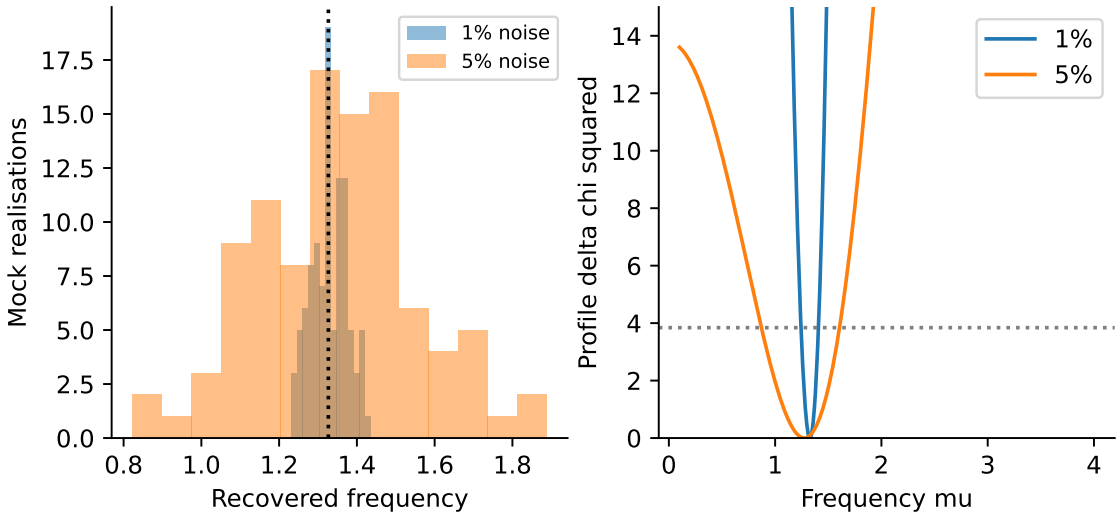



Figure 1: Mock recovered frequencies and example profile curves. Coverage measures the mock likelihood procedure, rather than calibrating deterministic numerical errors in the physical calculation.

For a no-clock state, frequency is not identified, so a naive fixed-frequency significance test would miss the frequency search. We calibrate the improvement over the two-term background on 100 null realisations and test it on another 100 independent nulls per noise level. The held-out false-alarm rates are 2% and 4%. With 100 trials, differences of a few percentage points are not decisive.

## 2 Physical clock fit and finite-window limitations

After the Step 8 pilot passes, we evaluate 50 physical triangles on $10^{-3} \leq r \leq 0.2$. Regulator extrapolation is repeated at every point. The numerical weights combine the local regulator change with the other pilot error-budget contributions, interpolated in $\ln r$. They are deterministic accuracy diagnostics and do not define an observational statistical confidence interval.

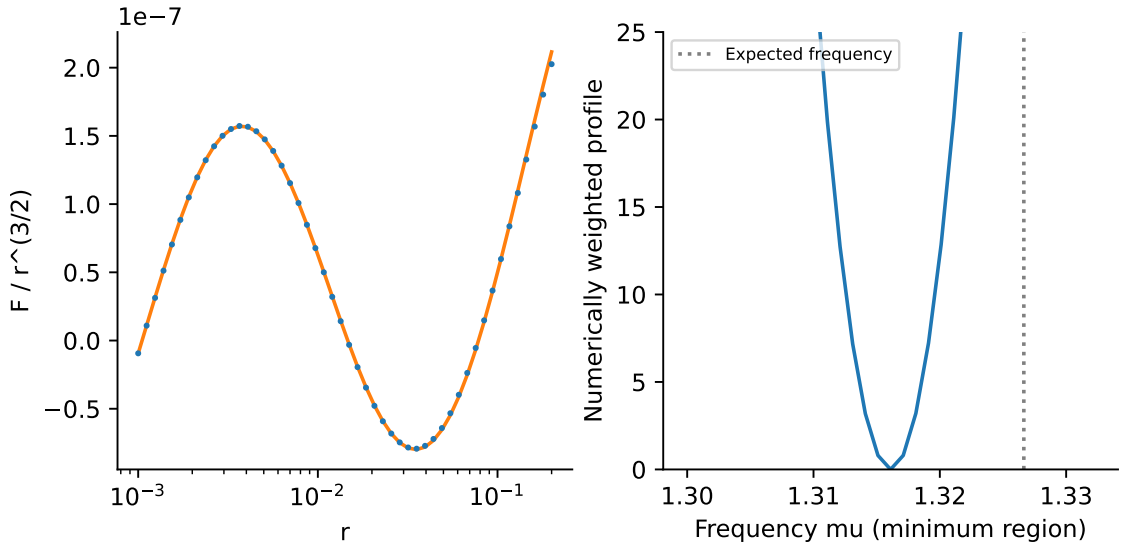



Figure 2: The physical bispectrum divided by $r^{3/2}$ and its numerical frequency profile. A narrow profile under one background choice does not include fit-model or window error.

The full-range leading fit has $\mu = 1.316103$ and an oscillatory coefficient magnitude $\sqrt{A^2 + B^2} = 1.389 \times 10^{-7}$. The maximum numerical error divided by the fitted clock envelope is 0.008524, below the 0.1 gate. However, only about 1.11 cycles are visible, making the background model consequential.

The late-time mass relation is

$$m/H = \sqrt{\mu^2 + 9/4 - g^2}. \tag{2}$$

Omitting $g^2$ is a weak-mixing approximation. If $g$ is unknown, the clock measures a mass–mixing combination.

## 3 Window and background stability

We vary the upper endpoint $r_{\max} = 0.05, 0.1, 0.2$, the lower endpoint $r_{\min} = 0.001, 0.003$, and add an $r^4$ analytic term as a background control. The resulting best fits are

| $r_{\min}$ | $r_{\max}$ | Extra $r^4$ | $\mu$ | $m/H$ |
|---|---|---|---|---|
| 0.001 | 0.05 | no | 1.32317 | 1.99769 |
| 0.001 | 0.05 | yes | 1.32544 | 1.9992 |
| 0.003 | 0.05 | no | 1.31765 | 1.99404 |
| 0.003 | 0.05 | yes | 1.32421 | 1.99838 |
| 0.001 | 0.1 | no | 1.32041 | 1.99587 |
| 0.001 | 0.1 | yes | 1.32445 | 1.99854 |
| 0.003 | 0.1 | no | 1.31637 | 1.9932 |
| 0.003 | 0.1 | yes | 1.32142 | 1.99654 |
| 0.001 | 0.2 | no | 1.3161 | 1.99302 |
| 0.001 | 0.2 | yes | 1.32196 | 1.99689 |
| 0.003 | 0.2 | no | 1.35637 | 2.01983 |
| 0.003 | 0.2 | yes | 1.31787 | 1.99419 |

The best-fit frequency envelope is 1.3161–1.35637, corresponding to a mass envelope 1.99302–2.01983. This is a sensitivity range across the specified analyses, not a confidence interval with demonstrated coverage. The expected $\mu = 1.326649916$ lies in this range. Restricting to the more squeezed $r \leq 0.05$ interval with the added analytic term produces a closer central estimate, while reducing the interval also weakens identifiability. We therefore report a resolved heavy-clock shape with model-dependent mass precision, rather than selecting the narrowest profile as the final error. The calibrated mock recovery and the physical numerical fit answer different questions.

## References

[1] N. Arkani-Hamed and J. Maldacena, *Cosmological Collider Physics*, arXiv:1503.08043.

**Code and supporting data:** step9 on GitHub.

## Contributions (equal)

| Member | Contribution |
|---|---|
| Priyansh Rawat | Frequency model and mass-conversion interpretation. |
| Samarth Bhatnagar | Mock/no-clock calibration and fit-window review. |
| Mayank Sinha | Profile fitting, physical shape scan and figures. |